Journal of Hydrology, 45 (1980) 169—184

© Elsevier Scientific Publishing Company, Amsterdam — Printed in The Netherlands

[2]

DISPERSIVITY AND VELOCITY RELATIONSHIP FROM LABORATORY AND FIELD EXPERIMENTS

D. KLOTZ, K.-P. SEILER, H. MOSER* and F. NEUMAIER

Institut für Radiohydrometrie der Gesellschaft für Strahlen- und Umweltforschung m.b.H., D-8042 Neuherberg—Munich (Federal Republic of Germany)

(Received March 5, 1979; revised and accepted June 29, 1979)

ABSTRACT

Klotz, D., Seiler, K.-P., Moser, H. and Neumaier, F., 1980. Dispersivity and velocity relationship from laboratory and field experiments. J. Hydrol., 45: 169—184.

Laboratory test results on longitudinal dispersion of soluble material in loose rocks with artificially composed or natural grain-size distributions are compared with one another and with results of field tests. In this way it can be demonstrated that it is quite possible to translate laboratory results to field conditions, if sedimentological properties of the loose materials are comparable, if bedding (geological interfaces) runs parallel to the flow direction and is hydraulically ineffective, and if distances between injection and detection are about less than 50 m.

Transverse dispersion is difficult to determine. At first approximation the width of tracer cloud from a line injection was determined and the aperture angle of the resultant cloud calculated. The laboratory and field experiments showed an aperture angle of about 5° on distances between 10 and 2500 m.

1. INTRODUCTION

The study of the behaviour of soluble material in groundwater is of utmost importance, particularly in connection with groundwater protection schemes. Apart from the interaction between the dissolved material and the rock and from molecular diffusion and in addition to the chemobiological degradation, hydrodynamic dispersion contributes essentially to the modification of concentration of soluble material in the aquifer.

During the past years, the theory of hydrodynamic dispersion in porous materials (e.g., loose rocks), based on statistical models (Scheidegger, 1963), has been developed for a series of boundary conditions (Harleman and Rumer, 1963; Bachmat and Bear, 1964; Lenda and Zuber, 1970; Fried, 1972, 1975; Hibsch and Kreft, 1979). However, a great number of laboratory and field tests, which have been carried out on the basis of most different practical objects, have been compared with one another only in some specific cases (Lenda and Zuber, 1970; Fried, 1972; Bertsch, 1978). In this connection, it was par-

*To whom correspondence should be addressed.

ticularly difficult to compare the results of laboratory tests with field tests, because of different hydraulic, hydrogeological boundary conditions and of scale parameters.

This study presents dispersion investigations in the laboratory which were confirmed through field tests. In this way one can create a better basis for laboratory testing methods to determine the dispersion conditions in natural aquifers and thus can limit field testing.

2. THEORETICAL AND EXPERIMENTAL PREREQUISITES

As a measure for the hydrodynamic dispersion, the dispersion coefficient D is usually used. This coefficient can be determined on the basis of temporal and local variation of the concentration of a conservative tracer. For the two-dimensional case, it may be defined by the following differential equation:

$$\partial c/\partial t = D_L\,(\partial^2 c/\partial x^2) + D_T\,(\partial^2 c/\partial y^2) - v_a\,(\partial c/\partial x) \qquad (1)$$

where

c = tracer concentration in the sampling fluid
v_a = average velocity as flow velocity of groundwater between two points on the same flow line
x, y = Cartesian coordinates (x-axis in the flow direction, and y-axis in the horizontal plane
t = time
D_L = longitudinal dispersion coefficient
D_T = transverse dispersion coefficient

The solution of eq.1 depends on the initial and boundary conditions, which should reflect the geometry of the field as well as the tracer injection mode and the way of measuring its concentration.

For a two-dimensional flow pattern in an isotropic and homogeneous porous medium with instantaneous tracer injection throughout the full thickness h (line injection), the initial and boundary conditions for the solution of eq. 1 are as follows:

boundary conditions:

$$c(x=0, y, t) = \frac{A}{n_e h v_a}\,\delta(y)\delta(t)$$

$$\lim_{x\to\infty} c(x, y, t) = 0; \qquad \lim_{y\to\pm\infty} c(x, y, t) = 0$$

initial condition:

$$c(x, y, t=0) = 0 \qquad \text{for } x > 0 \qquad \text{and} \qquad y \neq 0$$

where

A = the added tracer amount

n_e = the effective porosity
$\delta(y), \delta(t)$ = the Dirac delta functions for the y and t variables, respectively

For these boundary and initial conditions the solution of eq.1 is given by Lenda and Zuber (1970), and Hibsch and Kreft (1979) as:

$$c(x, y, t) = \frac{Ax}{4\pi n_e h v_a t^2 (D_L D_T)^{1/2}} \exp - \left[\frac{(x - v_a t)^2}{4D_L t} + \frac{y^2}{4D_T t} \right] \tag{2}$$

For an one-dimensional flow pattern, under corresponding initial and boundary conditions we obtain the following solution (Lenda and Zuber, 1970):

$$c(x, t) = \frac{Ax}{n_e F v_a t (4\pi D_L t)^{1/2}} \exp \frac{-(x - v_a t)^2}{4D_L t} \tag{3}$$

where F is the cross-section of the aquifer.

The magnitude of the dispersion coefficient depends on the velocity and on the sedimentological properties of the loose rock. A series of laboratory tests were carried out on artificially produced particle mixtures (Klotz, 1973; Klotz and Moser, 1974). Whereby it was found that D_L is related to the velocity v_a by the equation:

$$D_L = B v_a^A \tag{4}$$

B has the dimension of length, as long as $A = 1$.

The relation $B = D_L / v_a^A$ ("dispersivity") depends on the effective grain size (Fig.1) and on the uniformity coefficient (d_{60}/d_{10}) (Fig.2). The influence of the grain shape, the grain roughness and the packing as well as the water temperature are of minor importance. It must, however, be taken into account that dispersion is also influenced by geological interfaces and the distribution of sediment bodies in the direction of flow of the groundwater. Particularly on very short test distances (a few meters), this may adversely affect any comparison with laboratory tests and the relation between the dispersion and velocity because statistical homogeneity of porous media on the small scale is not yet given.

This is the reason for basing our laboratory and field tests on loose rock samples originating from the same accumulation areas; thus, these samples differ sedimentologically from each other only in a few parameters. In this way, we get an insight into the influence of special sedimentological properties on the dispersion and we may distinguish these from the influences of geological interfaces.

3. MEASURING ARRANGEMENTS AND PROCEDURES

3.1. Laboratory tests

The laboratory tests were carried out in one-dimensional column devices and in three-dimensional flow containers. Fig.3 is a schematic drawing of the

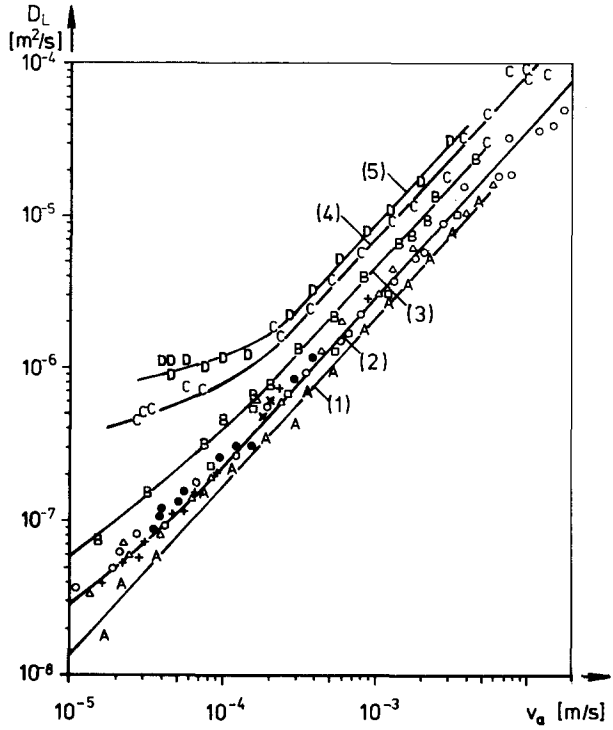

Fig.1. Longitudinal dispersion coefficient D_L in relation to distance velocity v_a for quartz sands and gravels (uniformity coefficient $U < 2$) of different grain size.
Tracers: *A*, *B*, *C*, *D* and ○ = $NH_4{}^{82}Br$; ● = $Na^{131}J$; + = ^{51}Cr-EDTA; △ = NaCl; □ = fluorescein (Uranine®); × = $^3H^1HO$.

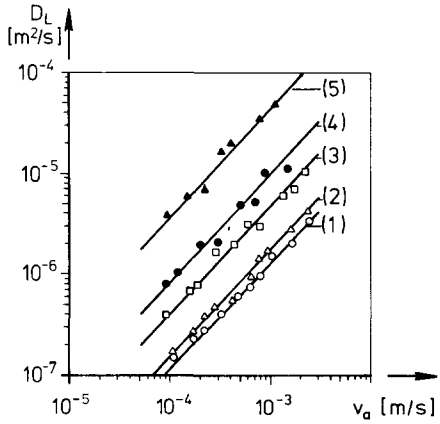

Fig.2. Longitudinal dispersion coefficient D_L in relation to distance velocity v_a for compacted sands of mean grain diameter d_{50} = 1 mm with different uniformity coefficients *U*. (*1*): $U = 1.1$; (*2*): $U = 2$; (*3*): $U = 5$; (*4*): $U = 10$; (*5*): $U = 20$.

column devices. The columns used were 250—4000 mm long and had a diameter of 10—500 mm. The injection device that was used guarantees that the tracer solution penetrates through the porous material in a uniform layer and throughout the full column section. As tracer elements, solutions of radioactive nuclides (NH_4 ^{82}Br, $Na^{131}J$, ^{51}Cr-EDTA, $^{3}H^{1}HO$), NaCl solutions and fluorescent dyes [e.g., fluorescein (Uranine®)] were used. We proved these tracers to be conservative in the loose rock materials tested in comparison with hydraulic parameters determined by classical methods (see Fig.1). The concentration was maintained on such a low level that no density convection appeared ($c < 10^{-1}$ mol/l H_2O for radioactive tracer and NaCl, $c < 10^{-7}$ kg/l H_2O for Uranine®).

The flow container used is schematically illustrated in Fig.4. The porous medium is delimited by filters against the stilling chambers. At the bottom and on one side-wall of the container piezometers have been provided to determine the hydraulic pressure distribution within the container.

To get a stable grain skeleton for laboratory tests, a procedure, according to which the porous material is successively built up within water, in layers of ~ 50 mm thickness, has proved to be an optimum solution. Thereby, the material is compacted with a round rod (Klotz, 1973).

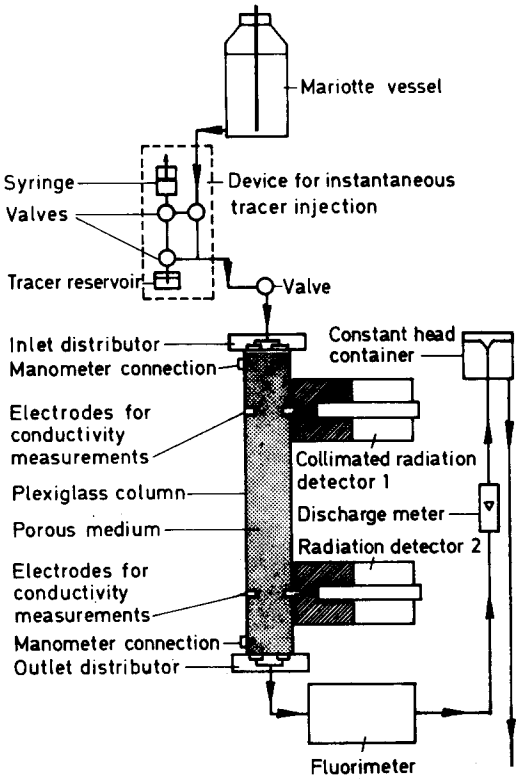

Fig.3. Schematic representation of the column setup to measure longitudinal dispersion.

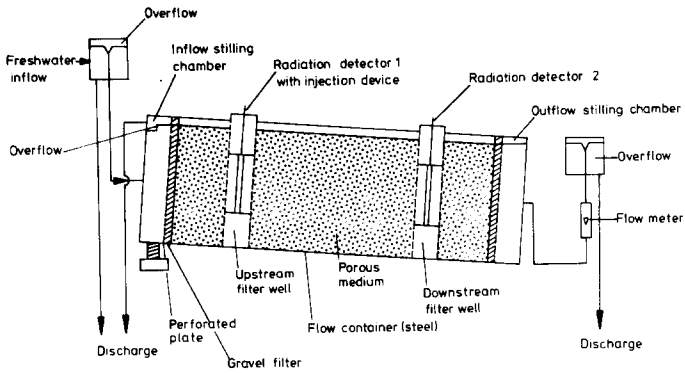

Fig.4. Schematic representation of the flow container to measure longitudinal dispersion.

3.2. *Field tests*

The field tests were carried out in fluvio-glacial gravels in a test field with eight observation wells arranged on a semi-circle around a pumping well (Fig.5). With the exception of observation well *10*, all drilled wells reach down to an impermeable layer at 13.8 m below land surface. The tracer tests were accomplished with and without groundwater extraction from pumping

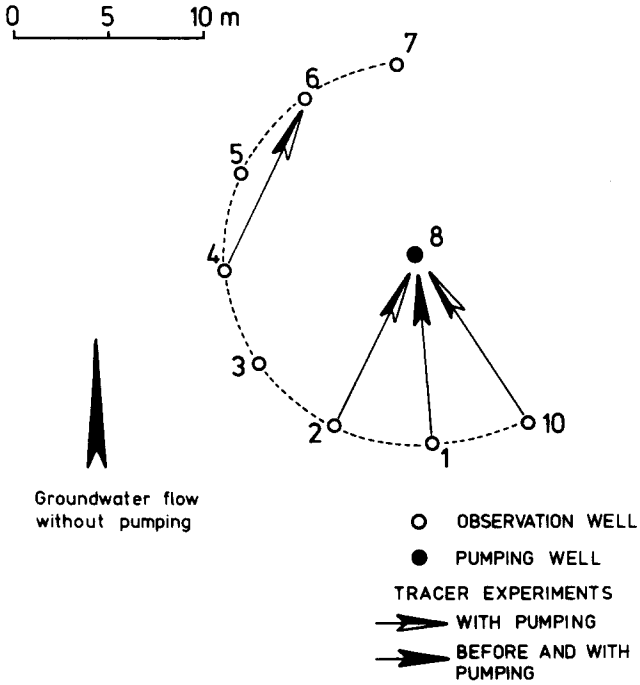

Fig.5. The pumping well and the surrounding observation wells on the Dornach test field near Munich.

well *8*. The addition of the tracer was effected alternately to wells *1*, *2*, *4* and *10*. In wells *1*, *2* and *4*, there was during all tests a vertical downward flow in the axis of the wells. Here, the tracer was added at the sinks of the vertical flow (at 13.8 m below land surface). Well *10* is open only between 8 and 9 m; hence, the tracer was added at this depth. In wells *6* and *8*, the tracer could be detected only at the depth where it had been added.

$NH_4{}^{82}Br$ and (for comparison) Uranine® have been used as tracers. The quantity of tracer material recovered was about 100% with groundwater extraction and about 80% without groundwater extraction in wells *6* and *8*.

In addition to the measuring results obtained on the test field, further tracer tests accomplished in Upper Bavaria have been taken into account in connection with the evaluation of the tests.

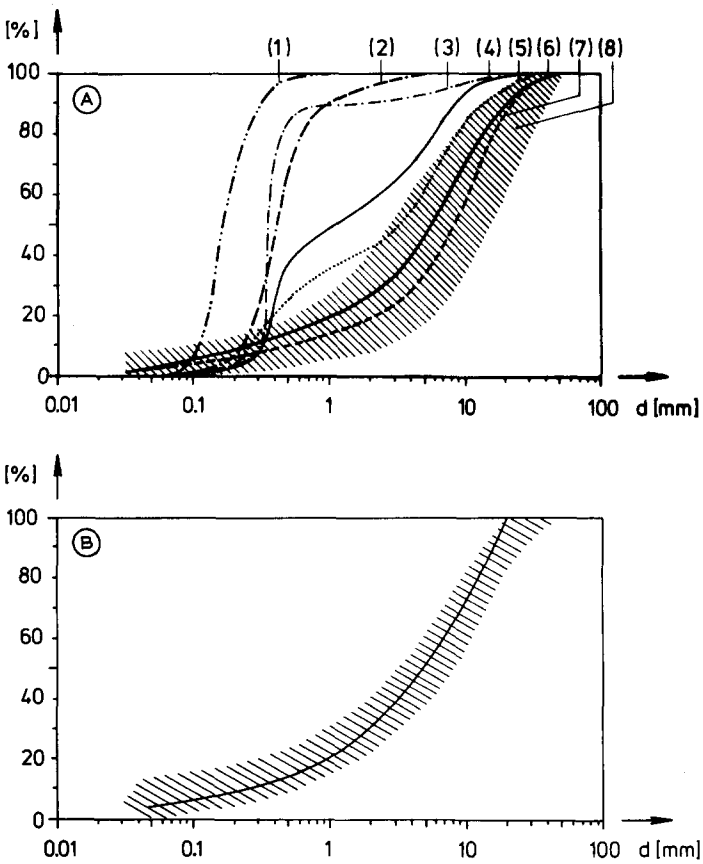

Fig. 6. A. Grain-size distribution curves of Quaternary aeolian sand (*1*) and river sand (*2*) from northern Germany; of Tertiary sands and gravels from Bavaria (*3*)—(*5*); and fluvioglacial gravels taken from the Dornach test field (*6*), (*7*); and grain-size range of fluvioglacial gravels taken from different sites in Bavaria (*8*).
B. Range of grain-size distributions for 110 samples from the Dornach test field.

3.3. *Loose rocks used as test material*

Fluvio-glacial gravel deposits cover regionally large areas in Upper Bavaria and are the main aquifers. The range of grain-size distribution of the gravels tested in the laboratory and that of in the field are shown in Fig.6. As a rule, the uniformity coefficient U of the fluvio-glacial gravels tested in the laboratory ($5 < U < 45$) is somewhat lower than that of the field tested samples ($10 < U < 150$). However, the range of grain-size distribution can be represented for all these materials by the grain-size distribution of a Fuller mixture having a maximum grain diameter of about 20 mm.

For the purposes of comparison, Fig.6 shows also the sieve curves of Tertiary loose rocks and Quaternary sands.

3.4. *Determination of the longitudinal and transverse dispersion coefficients*

The result of tracer tests is a concentration—time distribution at the measuring point x corresponding to eq.3. Accordingly, when measuring in the direction of flow, the longitudinal dispersion coefficient D_L will be obtained, approximately from the standard deviation σ of the concentration—time distribution, as per the following equation (Fried, 1972):

$$D_L = \sigma^2/2T \tag{5}$$

where T is the time for the center of gravity of the concentration—time distribution. In field testing, instead of the approximately symmetrical tracer transmission curve of eq.3, asymmetrical concentration—time distributions are frequently obtained. If one is able to analyse these asymmetrical concentration—time distributions, it will be possible to specify separately the different D_L-values. As a rule, the transverse dispersion coefficient D_T may be determined, according to eq.2, together with D_L, from the concentration—time distribution at two measuring points $s_1(x_1,y_1)$ and $s_2(x_2,y_2)$. A similar procedure has been proposed by Hibsch and Kreft (1979). Previous efforts to determine D_T in this way, on the basis of laboratory tests, have failed due to the fact that the transverse dispersion is very small in relation to the longitudinal dispersion. So far, the method has not been tried in connection with the results of field tests (see Section 4.4).

4. RESULTS

The results illustrated below originate from about 4000 tests carried out in column devices, about 300 tests in flow containers and about 20 tests carried out in the field. The results are illustrated in the form of graphs of eq. 4 in double-logarithmic scale.

The A-value in eq. 4 is for all laboratory and field tests in the range of $1.0 \leqslant A \leqslant 1.2$. This approximate linear correlation between D_L and v_a is not valid below a certain distance velocity, which depends on the grain size of the porous

medium (see Fig.1). This is to be attributed to additional mixing effects [e.g., molecular diffusion, density convection (Klotz, 1973; Moser and Klotz, 1974)].

The value B depends mainly on the grain size and on the uniformity coefficient (Table I) and determines as "dispersivity" the dependence of the dispersion on the sedimentological parameters.

TABLE I

Parameters A and B of the longitudinal dispersion in artificially produced grain mixtures (from Klotz, 1973; Klotz and Moser, 1974) and their lower limit referring to the distance velocity

Grain size (mm)	d_{50} (mm)	U	n	Measuring values, see Fig.	A	B	Lower limit for v_a (m/s)
0.4—0.7	0.61	1.55	0.31	1 curve (*1*)	1.09	$3.96 \cdot 10^{-3}$	10^{-5}
0.5—1.5	0.75	1.85	0.32	1 curve (*2*)	1.10	$5.78 \cdot 10^{-3}$	$8 \cdot 10^{-5}$
1—2	1.6	1.6	0.34	1 curve (*3*)	1.10	$8.80 \cdot 10^{-3}$	$1.5 \cdot 10^{-4}$
2—3	2.7	1.3	0.31	1 curve (*4*)	1.09	$1.30 \cdot 10^{-2}$	$2 \cdot 10^{-4}$
5—7	6.3	1.3	0.35	1 curve (*5*)	1.09	$1.67 \cdot 10^{-2}$	$3 \cdot 10^{-4}$
0.9—1.12	1.0	1.1	0.39	2 curve (*1*)	1.07	$2.15 \cdot 10^{-3}$	
0.5—2	1.0	2	0.33	2 curve (*2*)	1.08	$3.11 \cdot 10^{-3}$	
0.2—5	1.0	5	0.31	2 curve (*3*)	1.08	$8.30 \cdot 10^{-3}$	$5 \cdot 10^{-3}$
0.1—10	1.0	10	0.28	2 curve (*4*)	1.07	$1.63 \cdot 10^{-2}$	
0.05—20	1.0	20	0.28	2 curve (*5*)	1.07	$7.07 \cdot 10^{-2}$	

d_{50} grain diameter with 50% undersize; $U = d_{60}/d_{10}$ = uniformity coefficient; n = total porosity; A and B see eq.4; v_a = flow velocity (see eq.1).

4.1. Laboratory testing in column devices to determine the longitudinal dispersion in natural grain mixtures

Table II shows a list of natural grain mixtures which had been submitted for laboratory tests, together with their relative characteristics as derived from grain-size distribution (see Fig.6) as well as with their bulk porosity within the test configuration. Moreover, Table II shows the values A and B which have been derived from the measurements (as per Fig.7) of the dispersion coefficient D_L as a function of the velocity v_a in these grain mixtures.

The results so obtained confirm the findings about the longitudinal dispersion as derived from the measurements of artificially composed grain mixtures. Thus, the measuring results derived from the North German fine sands [curves (*1*) and (*2*) in Fig.7], show low values of the factor B, which can be attributed to the small grain size and low uniformity coefficient (Table II). In contrast to these findings, the measurements carried out with the Tertiary gravel-sands of Upper Bavaria [curves (*3*), (*4*) and (*5*) in Fig. 7] yielded higher

TABLE II

Parameters A and B of the longitudinal dispersion in natural grain mixtures

Type of sediments	Origin	d_{10} (mm)	d_{50} (mm)	U	n
Aeolian sands	North Germany	0.11	0.17	1.81	
Valley sands		0.23	0.42	1.96	
Tertiary gravel-sands	Upper Bavaria	0.32	0.35	1.13	0.25
		0.30	1.0	7.67	0.26
		0.25	3.0	18.8	0.25
10 fluvio-glacial gravels	Upper Bavaria	0.18—2.4	3.5—13	5.8—45	0.13—0.22
Fluvio-glacial gravels	samples from the test field	0.24	5.6	31.3	
		0.45	8.5	24.4	

d_{10}, d_{50} = grain diameter with 10% and 50% undersize, respectively; $U = d_{60}/d_{10}$ = uniformity coefficient; n = total porosity; A and B see eq. 4.

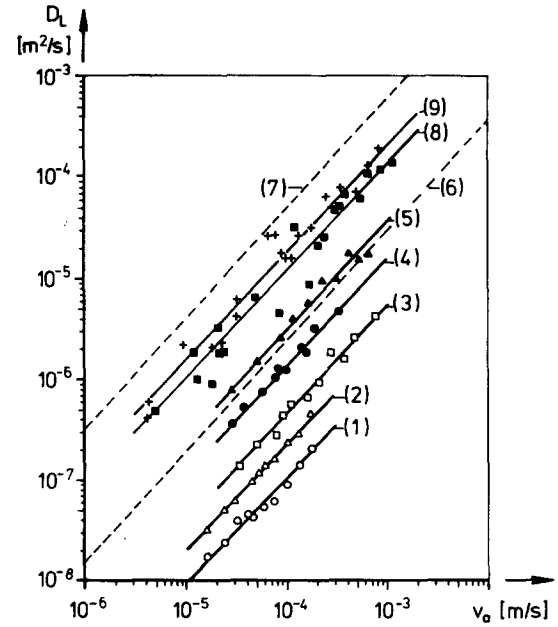

Fig. 7. Longitudinal dispersion coefficient D_L in relation to distance velocity v_a for different natural sands and gravels (see Fig. 6).
(*1*) = Quaternary aeolian sand; (*2*) = Quaternary river sand; (*3*)—(*5*) = Tertiary sands and gravels from Bavaria; (*6*), (*7*) = lower and upper limit in the relation between D_L and v_a for fluvio-glacial gravels from Bavaria; (*8*), (*9*) = fluvio-glacial gravels from the Dornach test field.

Measuring values, see Fig.	A	B
7 curve (*1*)	1.07	$2.05 \cdot 10^{-3}$
7 curve (*2*)	1.07	$4.64 \cdot 10^{-3}$
7 curve (*3*)	1.08	$1.01 \cdot 10^{-2}$
7 curve (*4*)	1.07	$2.59 \cdot 10^{-2}$
7 curve (*5*)	1.08	$6.72 \cdot 10^{-2}$
7 bond between curves (*6*) and (*7*)	1.09 for (*6*) 1.07 for (*7*)	$5.72 \cdot 10^{-2}$ for (*6*) $8.87 \cdot 10^{-1}$ for (*7*)
7 curve (*8*)	1.07	$4.47 \cdot 10^{-1}$
7 curve (*9*)	1.07	$2.35 \cdot 10^{-1}$

values of the factor B, since the test material was characterized by higher effective grain sizes and a higher uniformity coefficient. The largest dispersion as a function of a given velocity is shown by the fluvio-glacial gravels deposited in different areas of Bavaria. These show median grain diameters (d_{50}) of up to 13 mm and uniformity coefficients of up to 45. The relative values of the factor B fluctuate here between 0.057 and 0.89 m.

When comparing the characteristic values A and B, shown in Table I and derived from artificially produced grain mixtures, with the values as shown in Table II and derived from the natural grain mixtures, we note that in both cases D_L rises more or less linearly with v_a ($1.07 < A < 1.10$). Whenever the grain material shows more or less the same composition parameters, we also obtain more or less the same values for the factor B. This can be seen particularly well on the North German fine sands and the Tertiary gravel-sands. On the other hand, the fluvio-glacial gravels, which show a strongly variable composition in their grain-size distribution (see Fig. 6) can only be considered qualitatively in connection with this comparison. They are, however, compared in Section 4.2 with the results of field tests.

4.2. Field tests aimed at the determination of the longitudinal dispersion

The correlation between D_L and v_a, as obtained from measurements in the field test area, has been plotted in Fig.8. The different velocities have been obtained through variation of the pumping rate. They are typical of the velocities in fluvio-glacial gravels of Upper Bavaria.

As was determined in laboratory testing, field testing also yields a more or less linear correlation between D_L and v_a. The position of the flow path as well as the depth of the tracer addition does not affect the "dispersivity" D/v_a^A. After pumping started, values of D_L and v_a group around line *1*; two months later these values are about 25—30% lower and group around line *2* (Fig. 8).

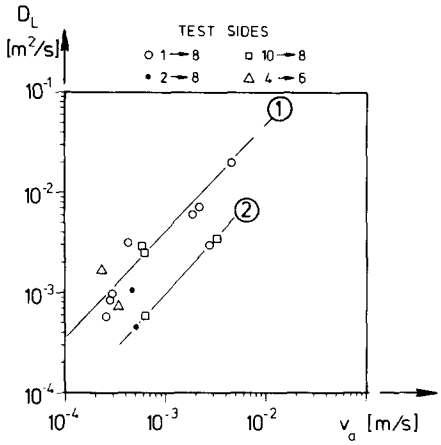

Fig.8. Relation between longitudinal dispersion D_L and distance velocity v_a by means of experiments of the Dornach test field. Line *1* = results obtained after pumping started; line *2* = results obtained two months after the commencement of pumping.

This is due to the fact that during the initial pumping test small amounts of sand were entrained with the test water. This changed the uniformity coefficient of the aquifer which affects very strongly the dispersion (see Sections 2 and 4.1).

The value A (1.05) of eq.4 corresponds to the value determined by laboratory tests on fluvio-glacial materials (Section 4.1). The value B was found to be 5 for line *1* and 1.9 for line *2* in Fig.8 and was 2—6 times higher than the laboratory values. This is due to the higher uniformity coefficient of the field material investigated (see Section 3.3). Preliminary results of measurements of longitudinal dispersion in other areas of fluvio-glacial gravel material, show approximately the same values for A and B for distances between injection and detection point from some meters to at least 50 m. Whether in this way regionally consistent parameters have been found with respect to the dispersion characteristics must be left undecided for the time being.

4.3. Laboratory tests in flow containers carried out to determine the longitudinal dispersion and first measurements to establish the transverse dispersion in column devices

In order to check and evaluate the results of the longitudinal dispersion in columns, a flow container with a cross-section of about 4 m² and a length of about 5 m was used to measure the velocity between two measuring gauges. The upstream gauge was labelled with an $NH_4{}^{82}Br$ solution, while in the downstream gauge, which was 3 m below the upstream gauge, the concentration—time distributions were measured. Fig.9 shows the longitudinal dispersion coefficient D_L as a function of the velocity v_a, as corrected by Klotz (1976), between the gauges. According to this, the longitudinal dispersion coefficient

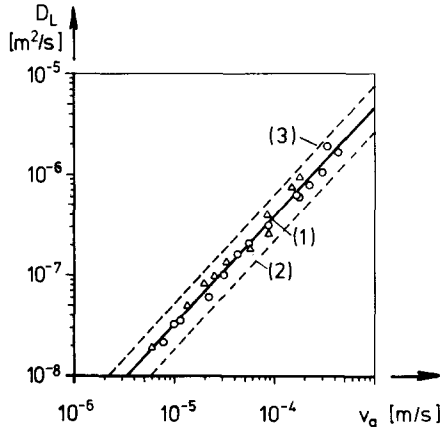

Fig.9. Longitudinal dispersion coefficient D_L in relation to distance velocity v_a for sand of grain size 0.5—1.5 mm (d_{50} = 0.8 ± 0.1 mm), uniformity coefficient U = 1.8 ± 0.3 (*1*) = flow container measurements between two plastic filter tubes without gravel filter (○) and between two steel filter tubes with gravel filter of grain size 2—3 mm (△); (*2*) and (*3*) = measurements in a column setup with total porosities of n = 0.42 and 0.43, respectively.

D_L is independent of the construction of the two gauges. The longitudinal dispersion coefficients from container experiments are the same as those determined in column devices on the same material (Table III). The small differences shown by the parameter B may be attributed to the different porosities.

As a first step to determine the transverse dispersion with the help of laboratory tests, measurements of the concentration—time distributions perpendicular to the flow direction were carried out in columns. This permits an evaluation of the width b of the tracer cloud. The two parameters, flow velocity and porous material, were varied. Within the low measuring accuracy, no dependence of the width b of the tracer cloud on the flow velocity could be ascertained. On the other hand, it was found that b increases with the increase of grain size of the porous medium. Across a flow extension of 115 cm, and with an injection of the tracer in sands with grain sizes of 0.1—0.4 mm and 0.2-

TABLE III

Parameters A and B of the longitudinal dispersion in a grain mixture of (d_{10} = 0.5 mm, d_{50} = 0.8 mm, U = 1.8), incorporated into different models

Mode of procedure	n	A	B
Flow container	undeterminable	1.08	$7.93 \cdot 10^{-3}$
Column device	0.42	1.08	$4.51 \cdot 10^{-3}$
Column device	0.34	1.08	$1.32 \cdot 10^{-3}$

d_{10}, d_{50} = grain diameter with 10% and 50% undersize, respectively; $U = d_{60}/d_{10}$ = uniformity coefficient; n = total porosity; A, B see eq. 4.

0.7 mm, a lateral dispersion of the tracer was recognized, with an aperture angle of 3°—5° (5.5 cm < b < 10.5 cm); whereas in sands of grain sizes of 0.5—1.5 mm and 1—2 mm, the aperture angle was 5°—7° (10.5 cm < b < 14.5 cm).

4.4. Results of measurements conducted in field tests to evaluate transverse dispersion

A direct determination of the transverse dispersion coefficients on the basis of tracer tests conducted in the field is difficult in most cases, as there generally are not enough observation wells to permit sufficiently precise measurement of the tracer concentration perpendicular to the flow direction.

As a semiquantitative measure of the transverse dispersion, we used the width of the tracer cloud. As the delimination of this width, we use the place where the detection limit of the tracer used is obtained [for fluorescein (Uranine®) this is $3 \cdot 10^{-12}$ kg/l (Behrens, 1971)].

Fig.10 shows the correlation between the width of the tracer cloud and the flow distance obtained by the measuring results. The results were obtained from our own as well as from other tracer tests conducted across flow distances between 10 and 2500 m (Cramer, 1972; Behrens, 1974). The flow sections are to be found in aquifers of fluvio-glacial gravels which are typical of the groundwater development areas of Upper Bavaria. Fig.10 shows that the width of the tracer cloud is smaller by about the factor 10 than the distance travelled by the tracer. By this order of magnitude, this is a qualitative confirmation

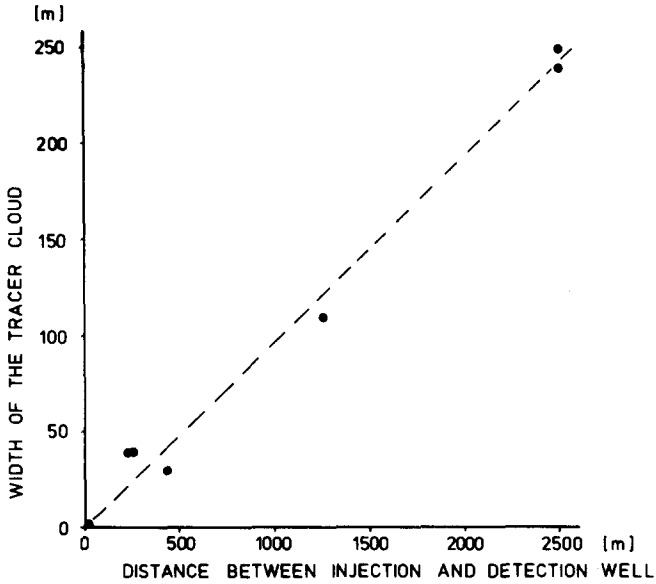

Fig.10. The width of the tracer cloud in relation to the length of flow path from labelling experiments on different test areas in Upper Bavaria.

of model tests (e.g., Harleman and Rumer, 1963; Li and Lai, 1966; Bruch and Street, 1967), where it had been found that D_T is 6—20 times smaller than D_L. Also the laboratory test results described in Section 4.3 produced the same aperture angle of the tracer cloud as has been found in the field tests.

The results show that groundwater labelling in loose rocks, effected over short observation distances and in areas without groundwater extraction, very often fails because the network of groundwater observation wells is generally too wide-spaced in relation to the width of the tracer cloud; hence, the tracer in many cases does not reach any groundwater observation well and therefore cannot be detected. Moreover, the results obtained agree with the evidence that within a series of wells arranged transversely to the groundwater flow direction, an individual well may be contaminated, without affecting the neighbouring wells, when the contamination originates from a point-source in the groundwater.

5. CONCLUSIONS

The test results demonstrate that it is quite possible up to distances of at least 50 m to transfer results of longitudinal dispersion of artificial grain-size distributions to natural grain-size distributions. This is best achieved with small uniformity coefficient.

Any transfer of dispersion parameters, obtained by laboratory testing, to the natural groundwater flow is only valid when the aquifer is found to have small variations in its sedimentological properties and when its sedimentological properties correspond to those of the porous material used in laboratory testing. Geological interfaces, which in many cases cause an additional hydrodynamic dispersion, generally cannot be simulated easily in laboratory testing.

ACKNOWLEDGEMENTS

We wish to extend our gratitude to Mrs. I. Heilmann and Messrs. M. Baumann, V. Müller, F. Oliv and H. Riedl, who have been a great help in our laboratory and field tests.

Moreover we are very much indebted to the Deutsche Forschungsgemeinschaft for its financial help (Mo 59/18, 20, 22-24, Ne 64/2-7) which has been used to finance a substantial part of the testing experiments.

REFERENCES

Bachmat, Y. and Bear, J., 1964, The general equations of hydrodynamic dispersion in homogeneous, isotropic porous medium. J. Geophys. Res., 69: 2561—2567.

Behrens, H., 1971. Untersuchungen zum quantitativen Nachweis von Fluoreszenzfarbstoffen bei ihrer Anwendung als hydrologische Markierungsstoffe. Geol. Bavarica, 64: 120—131.

Behrens, H., 1974. Untersuchungen im Bereich einer Mülldeponie. Inst. Radiohydrom. Ges. Strahlen-/Umweltforsch. Munich, Jahresber. 1973, pp. 92—102.

Bertsch, W., 1978. Die Koeffizienten der longitudinalen und transversalen hydrodynamischen Dispersion — ein Literaturüberblick. Dtsch. Gewässerkd. Mitt., 22: 36—46.
Bruch, J.C. and Street, R.L., 1967. Two-dimensional dispersion. J. Sanit. Eng. Div., Am. Soc. Civ. Eng., 6: 17—39.
Cramer, K., 1972. Ein Grundwasser-Markierungsversuch im Höhenkirchner Forst (Münchener Schotterebene, Oberbayern). Geol. Jahrb., C2: 265—274.
Fried, J.J., 1972. Études théorétiques et méthodologiques de la dispersion en milieu poreux naturel. Fac. Sci. Bordeaux, Bordeaux, No. 348.
Fried, J.J., 1975. Groundwater Pollution. Elsevier, Amsterdam, 330 pp.
Harleman, D.R. and Rumer, R.R., 1963. Longitudinal and lateral dispersion in an isotropic porous medium. J. Fluid Mech., 16: 385—394.
Hibsch, G. and Kreft, A., 1979. Interpretation of multi-well tracer experiments. J. Hydraul. Div. Proc. Am. Soc. Civ. Eng., 105 (in print).
Klotz, D., 1973. Untersuchungen zur Dispersion in porösen Medien. Z. Dtsch. Geol. Ges., 124: 523—533.
Klotz, D., 1976. Korrektur der Abstandsgeschwindigkeit bei Messungen zwischen zwei Pegeln. — Inst. Radiohydrom. Ges. Strahlen-/Umweltforsch., Munich, Jahresber. 1975, pp. 39—41.
Klotz, D. and Moser, H., 1974. Hydrodynamic dispersion as aquifer characteristic. In: Isotope Techniques in Groundwater Hydrology 1974, Vol. II, IAEA, Vienna, pp. 341—355.
Lenda, A. and Zuber, A., 1970. Tracer dispersion in groundwater experiments. In: Isotope Hydrology, IAEA, Vienna, pp. 619—641.
Li, W.H. and Lai, F.H., 1966. Experiments on lateral dispersion in porous media. J. Hydraul. Div., Proc. Am. Soc. Civ. Eng., 92: 141—149.
Scheidegger, A.E., 1963. Hydrodynamics in porous media. In: S. Flügge (Editor), Handbuch der Physik, 8/2, Springer, Heidelberg, pp. 625—662.